

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Lubricating Oil Cooler Pressure Test Kit, Part Number 3823876

Additional Service Items

- Brush
- Shop air
- Cleaning tank
- Screwdriver
- Gasket scraper

Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

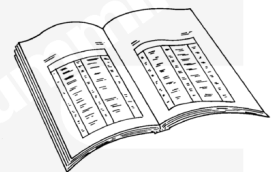
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

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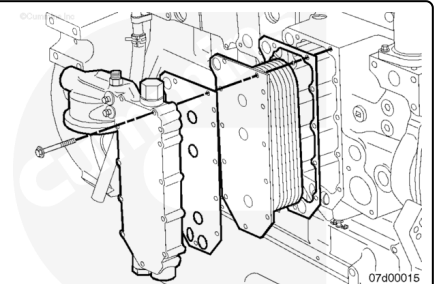
State and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html>)
- Clean the area around the lubricating oil cooler cover.
- Remove the coolant filter head. Refer to Procedure 008-007 in Section 8.
(</qs3/pubsys2/xml/en/procedures/100/100-008-007.html>)
- Remove the lubricating oil filter. Refer to Procedure 007-013 in Section 7.
(</qs3/pubsys2/xml/en/procedures/41/41-007-013-tr.html>)
- Remove the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10.
(</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)

Remove

Remove the lubricating oil cooler cover, gaskets, and cooler element.

Discard the gaskets.



Clean and Inspect for Reuse

⚠ WARNING ⚠

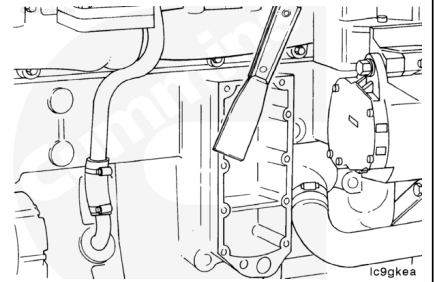
When using solvent, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Use a solvent that will not harm copper to clean the oil cooler elements.

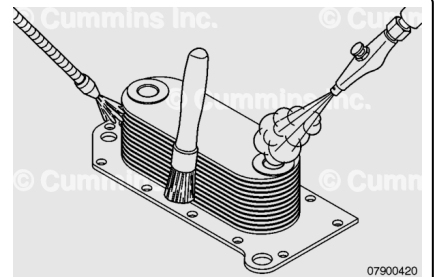
Clean the sealing surfaces of the cylinder block and the oil cooler cover.

Note : Replace the lubricating oil cooler if any debris is found or if the engine has had a debris-generating failure.



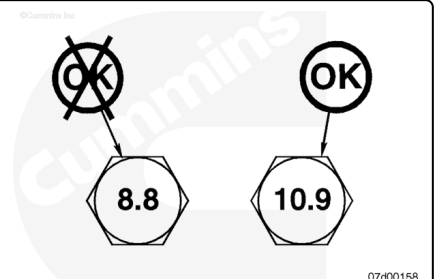
⚠ CAUTION ⚠

Do not reuse an oil cooler core after a debris related engine malfunction since there is no practical method for cleaning the cooler core. Metal particles which can circulate through the lubricating oil system can remain in the cooler core and cause engine damage.



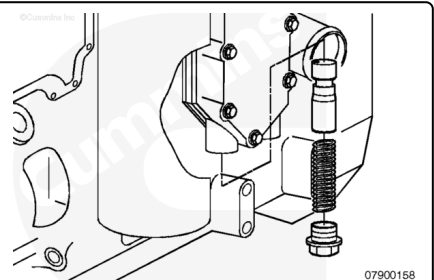
Inspect the capscrews. If they are Grade 10.9 or higher, they can be reused.

If the capscrews are lower than Grade 10.9, they **must** be replaced with Grade 10.9 or higher capscrews.



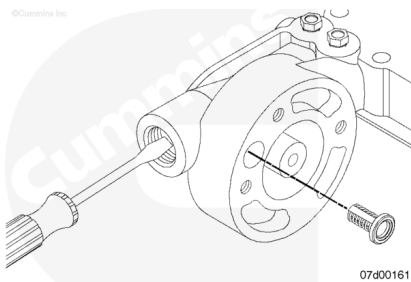
If any debris is suspected to have gone through the engine, or if troubleshooting a lubricating oil pressure issue, remove and inspect the lubricating oil pressure regulator located in the lubricating oil cooler cover. Replace if necessary. Refer to Procedure 007-029 in Section 7.

(/qs3/pubsys2/xml/en/procedures/100/100-007-029.html)



If any debris is suspected to have gone through the engine, inspect the oil filter bypass valve located in the lubricating cooler cover. Make sure the valve is fully seated and opens and closes freely. Replace if necessary.

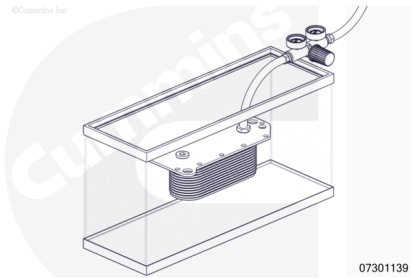
The bypass valve requires a 345 kPa [50 psi] pressure differential to open.



Leak Test

Use lubricating oil cooler pressure test kit, Part Number 3823876, to pressure-test the lubricating oil cooler element.

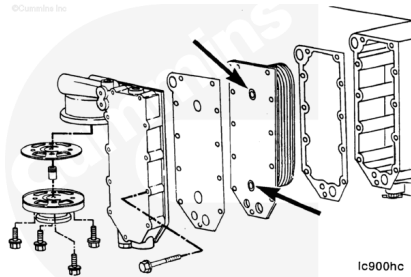
Air Pressure Test		
kpa		psi
449	MIN	65
518	MAX	75



Install

Note : Make sure to remove the shipping plugs from a new element.

Assemble the lubricating oil cooler gaskets, element, and cooler cover. Use new gaskets.



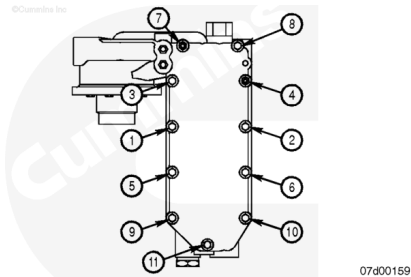
Tighten the capscrews in the sequence shown in the illustration, in two steps.

Torque Value:

1. 15 n•m [11 ft-lb]
2. 33 n•m [24 ft-lb]

Retorque capscrews 1 and 2 using the torque value listed below:

Torque Value: 33 n•m [24 ft-lb]



Finishing Steps

- Install the turbocharger oil supply line. Refer to Procedure 010-046 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>)
- Install the lubricating oil filter. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/41/41-007-013-tr.html>)
- Install the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-007.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html>)
- Operate the engine and check for leaks.
- Shut the engine OFF and check the coolant and lubricating oil levels.

